

TALENTFLOW

Project 3 • Blueprint v1.0

AI-Assisted Recruitment & Applicant Management Platform

A production-oriented flagship project for managing the complete hiring lifecycle — from job creation to hiring.

PROJECT 03

FULL-STACK

FLAGSHIP

v1.0

Product Vision

TalentFlow is a modern recruitment management platform that helps companies manage the complete hiring lifecycle — creating jobs, receiving applications, screening candidates, scheduling interviews, evaluating applicants, making offers, and tracking hiring performance.

Why Project 3 Is Different

Project 2 — JobFlow

Job seeker focused

"Where did I apply? What is my application status?"

Project 3 — TalentFlow

Company & recruiter focused

"Who applied? Who should we interview? Who should we hire?"

1. Core Users & Responsibilities

■ Candidate	Create profile, upload resume, browse/search jobs, save jobs, apply, track applications, view interviews, receive notifications.
■ Recruiter	Manage company, create jobs, review applicants, move candidates through stages, schedule interviews, add feedback, make offers, view analytics.
■■ Admin	Manage users/companies/jobs, monitor applications, view platform analytics, audit activity, and handle flagged content.

2. Core Recruitment Workflow

1 Create Job	2 Publish Job	3 Applications
4 Screening	5 Shortlisted	6 Interview
7 Evaluation	8 Offer	9 Hired

3. Feature Map

Authentication & RBAC	Candidate, recruiter, admin roles; secure login and permissions.
Job Management	Create, edit, publish, close, search and filter jobs.
Applications	Apply, review, manage status, prevent duplicate applications.
Candidate Pipeline	Kanban workflow with stage transitions and status history.
Resume Management	Secure upload, candidate profile, skills and experience.
Interviews	Scheduling, interview mode, meeting details and feedback.
Notifications	In-app events, application updates, interview reminders and email.
Analytics	Hiring funnel, application trends, conversion and time-to-hire metrics.
Intelligent Matching	Resume/job skill matching and candidate insights.
Audit & Security	Audit logs, validation, secure files, API protection and production settings.

4. Target System Architecture

5. Technology Stack

Frontend	React • JavaScript • Modern CSS • Charts • Drag & Drop
Backend	Python • Django • Django REST Framework
Database	PostgreSQL
Async	Celery • Redis
Infrastructure	Docker • Nginx • Gunicorn
Engineering	Git • GitHub • Testing • API documentation • CI/CD
Intelligent Features	Resume parsing • Skill extraction • Job/resume matching

6. Initial Domain Model

Conceptual Database

User → CandidateProfile / RecruiterProfile → Company → Job → Application

Application → StatusHistory / Interview / Offer

Supporting entities → Notification • SavedJob • Skill • JobSkill • CandidateSkill • InterviewFeedback • AuditLog

We will convert this conceptual model into a formal ER diagram before creating Django models.

7. UI / UX Direction

Recruiter Dashboard

SaaS-style sidebar, KPI cards, analytics, recent applications and hiring funnel.

Candidate Kanban

Applied → Screening → Shortlisted → Interview → Offer → Hired.

Responsive Design

Desktop-first productivity UI that remains usable on tablet and mobile.

Candidate Dashboard

Application timeline, upcoming interviews, saved jobs and notifications.

Job Detail

Clean job information, skills, salary/location, company details and application action.

Visual Language

Modern, clean, professional, with restrained cards, badges, charts and micro-interactions.

8. Security & Quality Principles

Production-minded engineering

- ✓ Secure authentication and role-based authorization
- ✓ CSRF protection and strict input validation
- ✓ Secure resume/file validation and storage
- ✓ API authentication and permission enforcement
- ✓ Environment variables for secrets and production configuration
- ✓ Audit logs for important recruitment actions
- ✓ Automated unit/API tests
- ✓ Production error handling, logging and monitoring

9. Development Roadmap

01	Foundation	Repository • Django • React • PostgreSQL • environment setup
02	Authentication	Registration • login • roles • permissions • profiles
03	Jobs	Create • publish • browse • search • filters
04	Applications	Apply • candidate profiles • resume • status workflow
05	Pipeline	Kanban • stage transitions • status history • evaluation
06	Interviews	Scheduling • feedback • candidate/recruiter notifications
07	Analytics	KPIs • funnel • charts • hiring metrics
08	Notifications	In-app • email • Celery • Redis
09	Intelligence	Resume parsing • skill extraction • matching
10	Engineering	Testing • API docs • security • logging
11	Deployment	Docker • production config • CI/CD • deployment

12	Portfolio	Architecture • ERD • flowcharts • screenshots • README • demo
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10. Progressive Release Strategy

MVP	Authentication + roles + companies + jobs + applications + candidate profiles + recruitment pipeline
V2	Interviews + notifications + analytics + resume management
V3	Celery + Redis + email + advanced analytics + audit logs
V4	AI-assisted resume parsing + job matching + Docker + CI/CD + production deployment

11. Flagship Project Standard

The rule for TalentFlow

Don't add technology just to say we used it.

Every component must solve a real product problem. Redis, Celery, AI, React, Docker, CI/CD and advanced analytics will be introduced where they improve the system — not merely increase the tech stack.

12. Definition of Success

TalentFlow is complete when...

- ✓ A recruiter can manage a real hiring pipeline end-to-end.
- ✓ A candidate can apply and transparently track progress.
- ✓ The system supports multiple roles with proper authorization.
- ✓ The frontend and backend communicate through a documented API.
- ✓ Background jobs and notifications work reliably.
- ✓ Analytics provide meaningful recruitment insights.
- ✓ Intelligent matching adds useful value without becoming a dependency.
- ✓ The project is tested, containerized, deployable, documented and portfolio-ready.

NEXT MILESTONE

System Architecture → User Flows → ER Diagram → Final Tech Decisions → Repository Structure

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